Belief sharing lowers free energy at the project-pool corner

With the standard-Bayes client limit and project-pool recovery identity pinned exactly (sec. 7.1), the study suite opens one step away from them. The first study's estimand is the colony's mean variational free energy under two communication conditions; the design is a categorical source-mechanism analogue of the colony belief-sharing result (Friston et al. 2024), implemented at the stated categorical recovery limits. A colony of 7 sentinels each observe the same hidden creature location through an independent noisy sensor (acuity 0.55) and form a one-step variational posterior. When the colony runs one federated belief-sharing round — the standard log-linear pool, which is the `robustness=0` corner of the aggregation identity eq. 7 proven in Theorem 5 — each agent's posterior moves toward the cross-agent consensus via the belief-sharing round eq. 10. Under the theorem's shared-support, posterior-log-potential, and fixed-weight assumptions, that pool specializes Eq. 7's message-combination term, not the complete source protocol; when held incommunicado, agents keep their private posteriors. Scoring each belief against the colony's joint evidence yields the "two heads are better than one" reduction, reported here as earned quantities rather than asserted:

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- Mean variational free energy, **communicating**: $\bar{F}_{\text{share}} = 13.2190$ (across-seed 95% bootstrap CI $[12.9656, 13.4685]$ over $n = 480$ seeds). The single illustrative seed-0 run has its own colony mean 15.8115, with a per-agent 95% bootstrap CI of $[15.6177, 16.0647]$ over its $n = 7$ agents — that interval characterizes the displayed run's per-agent spread, not the across-seed mean
- Mean variational free energy, **incommunicado**: $\bar{F}_{\text{solo}} = 16.5298$
- Free-energy reduction from sharing: $\Delta\bar{F} = 3.3109$ (communicating is strictly lower)

The across-seed colony means above are computed over $n = 480$ seeds. The communicating colony also reaches higher mean true-state accuracy (0.7302) and lower mean surprise (0.3775) than its members reach alone. Sharing pulls each private posterior toward the joint minimizer, so the mean free energy when communicating sits below the incommunicado value.

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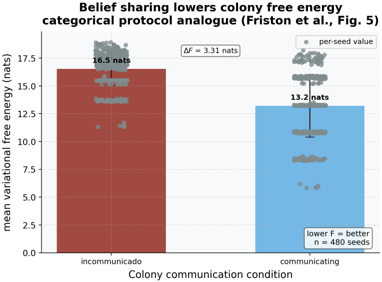

Figure 1: Mean variational free energy of the sentinel colony. Source relation: source-mechanism analogue to the belief-sharing mechanism in Friston et al. (2024), Fig. 5; estimand: colony-mean variational free energy in nats; uncertainty: across-seed spread over independent seeds. The sentinel colony (7 agents, acuity 0.55) under two communication conditions. x-axis: condition (incommunicado vs. communicating — one standard belief-sharing round), plotted in that order; y-axis: colony-mean variational free energy in nats. Bars show the mean free energy averaged across $n = 480$ independent random seeds, with whiskers marking $\pm$one across-seed standard deviation and grey points overlaying the individual per-seed values. The communicating bar is strictly lower than the incommunicado bar, with $\Delta\bar{F} = 3.3109$ nats — the quantitative “two heads are better than one” result. Each seed is a fully deterministic run; the whiskers are the across-seed spread, not a bootstrap or resampling interval.

fig. 1 reports the headline gap; fig. 2 shows the per-agent mechanism behind it.

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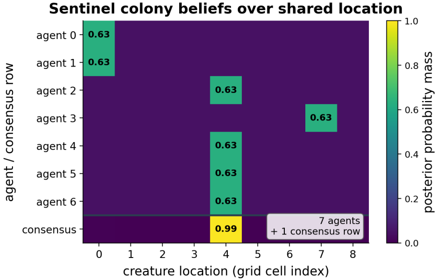

Figure 2: Single-panel belief heatmap over the hidden creature location. Source relation: original project diagnostic supporting the Study 1 analogue; estimand: posterior probability mass by hidden state; uncertainty: deterministic single-seed display. The hidden creature location (7 sentinels, acuity 0.55, seed 0). x-axis: hidden-state grid cell (creature location, 9 cells); rows: the 7 individual agents' private posteriors (one row per agent, dominant cell annotated), plus a bottom consensus row — separated by the divider line — holding the federated consensus fused from those posteriors by the cavity-exclusion round defined in the methods. Each private posterior concentrates only moderately on the cell its noisy observation suggests; the consensus row concentrates far more sharply on the true location, and that after-sharing concentration is the mechanism behind the free-energy reduction reported by the free-energy comparison. All cell values are deterministic posterior probabilities for the single displayed seed; no error band is applicable.

Three robustness axes remain distinct in the results I

Before the contaminated studies, we fix the honesty boundary that governs every robustness claim in this paper, because the belief-sharing round above is exactly where the axes diverge once contamination is introduced. The robust extension of belief sharing lives on **three distinct axes** that must not be conflated.

The **per-agent axis** is the generalized-Bayes update each sentinel runs locally: the β-loss and robust cross-entropy clients of eq. 4 and eq. 5. This axis follows the FedGVI objective (Mildner et al. 2025) and carries the cited bounded-influence result only under that theorem's matching assumptions. The recovery limits of sec. 7.1 show the per-agent update reduces to standard Bayes; the separately stated server identity returns the project log-linear pool.

The **server-side axis** is the `robust_aggregate` divergence-reweighting that down-weights agents at pooling time. This is a complementary *heuristic*. Its only proven property is the naive-recovery limit of Theorem 5 — at zero robustness it equals the standard log-linear pool (eq. 7) — and it carries **no** bounded-influence guarantee. No figure, table, or sentence in sec. 7.5 or sec. 7.6 grants the server-side heuristic the guarantee that belongs to the per-agent axis. The contaminated sweep that follows reports the axes side by side and labels which is which at every step.

Three robustness axes remain distinct in the results I

The **variational server axis** is `variational_aggregate`. It is also server-side, but it is not the same claim as the sharp heuristic: it descends the stated aggregation free energy (eq. 8), carries the derived effective-weight bound proved in the supplement, and pays for that property with a conservative maximum-entropy bias. The contaminated sweep therefore reports behavior for the sharp heuristic and cites the variational rule only where the objective-backed guarantee is actually in force.